

Prevalence of steatotic liver disease, MASLD, MetALD and significant fibrosis in people with HIV in the United States.

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Metabolic dysfunction-associated steatotic liver disease (MASLD) has recently been proposed as a replacement term for NAFLD. To assess the effects of this new nomenclature on the prevalence and distribution of different SLD categories in people with HIV (PWH) and identified factors associated with MASLD and clinically significant fibrosis (CSF). PWH were prospectively enrolled from 9 US centres and underwent clinical evaluation and vibration-controlled transient elastography for controlled attenuation parameter (CAP) and liver stiffness measurement (LSM). SLD was defined as CAP ≥ 263 dB/m, CSF as LSM of ≥ 8 kPa, and advanced fibrosis (AF) as LSM ≥ 12 kPa. The prevalence of SLD, MASLD, metabolic dysfunction and alcohol-associated liver disease (MetALD), ALD, cryptogenic (cSLD), CSF and AF were determined. Uni- and multivariate logistic regression models were used to assess factors associated with MASLD and CSF risk. Of 1065 participants, 74% were male, mean (SD) age 51.6 ± 11.9 years, 46% non-Hispanic Black and 74% with undetectable HIV RNA. The prevalence of SLD was 52%, MASLD 39%, MetALD 10%, ALD 3%, CSF 15% and AF 4%. Only 0.6% had cSLD. Black race was protective whereas obesity, ALT and AST levels were associated with increased risk of MASLD and CSF in MASLD. HIV or antiretroviral therapy did not affect MASLD risk. MASLD and MetALD are the dominant causes of SLD in PWH, affecting almost half. Application of the new nomenclature resulted in minimal change in the proportion of patients with MASLD who would have been diagnosed previously with NAFLD.

Emerging role of natural lipophagy modulators in metabolic dysfunction-associated steatotic liver disease.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), previously known as non-alcoholic fatty liver disease (NAFLD), is a seriously increasing liver disorder affecting nearly 32% of adults globally. Hepatic triglycerides (TG) accumulation is the hallmark of MASLD, which results from dysregulated lipid and fatty acid uptake, increased de novo lipogenesis (DNL), and decreased lipid removal. More recently, selective autophagy of lipid droplets (LDs), termed lipophagy, has emerged to be closely associated with disrupted hepatic lipid homeostasis. Recent studies have indicated that a series of natural products have shown promise as an alternative approach in attenuating MASLD via regulating lipophagy in vivo and in vitro. Therefore, lipophagy could be a new approach for natural products to be used to improve MASLD. This article aims to provide a comprehensive overview on the interrelationship between dysregulated lipid metabolism, lipophagy, and MASLD pathogenesis. In addition, the role of some natural products as lipophagy modulators and their impact on MASLD will be discussed.

Exploring the Multifaceted Landscape of MASLD: A Comprehensive Synthesis of Recent Studies, from Pathophysiology to Organoids and Beyond.

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Non-alcoholic fatty liver disease (NAFLD) is a widespread contributor to chronic liver disease globally. A recent consensus on renaming liver disease was established, and metabolic dysfunction-associated

steatotic liver disease, MASLD, was chosen as the replacement for NAFLD. The disease's range extends from the less severe MASLD, previously known as non-alcoholic fatty liver (NAFL), to the more intense metabolic dysfunction-associated steatohepatitis (MASH), previously known as non-alcoholic steatohepatitis (NASH), characterized by inflammation and apoptosis. This research project endeavors to comprehensively synthesize the most recent studies on MASLD, encompassing a wide spectrum of topics such as pathophysiology, risk factors, dietary influences, lifestyle management, genetics, epigenetics, therapeutic approaches, and the prospective trajectory of MASLD, particularly exploring its connection with organoids.

Epidemiology, screening, and co-management of type 2 diabetes mellitus and metabolic dysfunction-associated steatotic liver disease.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), previously known as NAFLD, is increasingly recognized as a prevalent global burden. Type 2 diabetes mellitus (T2DM), another important metabolic disease, is considered a major contributor to the development of MASLD. MASLD and T2DM have a strong association with each other due to shared pathogenic mechanisms. The co-existence of the 2 diseases increases the risk of liver-related adverse outcomes and imposes a heavier burden on extrahepatic outcomes, representing a substantial public health issue. Effective assessment and management of T2DM combined with MASLD necessitate a multidisciplinary approach. The emergence of numerous RCTs has shed light on the treatment of T2DM combined with MASLD. This review uncovers the epidemiology of the intertwined T2DM and MASLD, offers insights into the evaluation of hepatic fibrosis in patients with T2DM, glucose monitoring in the MASLD population, and provides comprehensive co-management strategies for addressing both diseases.

Visceral Fat and Diabetes: Associations With Liver Fibrosis in Metabolic Dysfunction-Associated Steatotic Liver Disease.

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The prevalence of metabolic dysfunction-associated steatotic liver disease (MASLD), previously known as non-alcoholic fatty liver disease, is increasing globally. Noninvasive methods, such as bioelectrical impedance analysis (BIA), which measures body composition, including visceral fat, are gaining interest in evaluating MASLD patients. Our study aimed to identify factors associated with significant liver fibrosis, compare noninvasive scores, and highlight the importance of visceral fat measurement using BIA. MASLD patients seen in our out-patient department underwent comprehensive evaluations, including liver stiffness using transient elastography, body composition analysis using BIA, and metabolic measurements. Significant fibrosis was defined as a liver stiffness measurement of ≥ 8.2 kPa. Using multivariate analysis, we identified factors associated with significant liver fibrosis and compared four noninvasive scores with a novel diabetes-visceral fat 15 (DVF15) score. We analyzed data from 609 MASLD patients seen between February 2022 and March 2023. The median age was 43 years (81% male). Among these, 78 (13%) had significant fibrosis. Patients with significant fibrosis had higher rates of type 2 diabetes (41% vs 21%, $P = .015$) as associated with significant liver fibrosis. Additionally, the DVF15 score, combining these factors, showed a modest area under the receiver operating characteristic

curve of 0.664 (P 15% should prompt clinicians to check for significant fibrosis in MASLD patients. Further research is warranted to validate our findings and evaluate the utility of the DVF15 score in larger cohorts and diverse populations.

MASLD and the Development of HCC: Pathogenesis and Therapeutic Challenges.

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Metabolic-dysfunction-associated steatotic liver disease (MASLD, previously known as non-alcoholic fatty liver disease (NAFLD)) represents a rapidly increasing cause of chronic liver disease and hepatocellular carcinoma (HCC), mirroring increasing rates of obesity and metabolic syndrome in the Western world. MASLD-HCC can develop at an earlier stage of fibrosis compared to other causes of chronic liver disease, presenting challenges in how to risk-stratify patients to set up effective screening programmes. Therapeutic decision making for MASLD-HCC is also complicated by medical comorbidities and disease presentation at a later stage. The response to treatment, particularly immune checkpoint inhibitors, may vary by the aetiology of the disease, and, in the future, patient stratification will be key to optimizing the therapeutic pathways.

Effect of N-acetyl cysteine in children with metabolic dysfunction-associated steatotic liver disease-A pilot study.

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Prevalence of metabolic dysfunction-associated steatotic liver disease (MASLD), previously known as nonalcoholic fatty liver disease (NAFLD), and its sequelae of more severe forms such as metabolic dysfunction-associated steatohepatitis (MASH) is rapidly increasing in children with the rise in obesity. Successful and sustainable treatments for MASLD are lacking in children. We determined the therapeutic effect of N-acetyl cysteine (NAC) on biomarkers of oxidative stress, inflammation and insulin resistance (IR), liver enzymes, liver fat fraction (LFF) and liver stiffness (LS) in children with obesity and biopsy-confirmed MASLD. Thirteen children (n = 13; age: 13.6 ± 2.8 years; NAS score >2) underwent a double-blind, placebo-controlled trial of NAC (either 600 or 1200 mg NAC/day) or placebo for 16 weeks. Measurements included LFF (magnetic resonance imaging), LS (ultrasound elastography), and body composition. Erythrocyte glutathione (GSH), liver enzymes, insulin, glucose, adiponectin, high-sensitivity c-reactive protein (hs-CRP), and interleukin-6 (IL-6) were also measured. homeostasis model assessment for insulin resistance (HOMA-IR) was calculated. Sixteen-week NAC treatment improved (baseline adjusted between-group $p < .05$ for all) markers of inflammation (IL-6 and hs-CRP), oxidative stress (GSH), and IR (HOMA-IR) and reduced liver enzymes, LFF and LS. Body weight and body composition did not show beneficial changes. Sixteen-week NAC treatment was well tolerated in children with obesity and MASLD and led to improvements in oxidative stress, inflammation and IR and liver outcomes. The results from this pilot study support further investigation of NAC as a therapeutic agent in children with MASLD.

Management of cardiovascular risk in patients with metabolic dysfunction-associated steatotic liver disease.

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The novel term Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD) is proposed to replace non-alcoholic fatty liver disease (NAFLD) to highlight the close association with the metabolic syndrome. MASLD encompasses patients with liver steatosis and at least one of five cardiometabolic risk factors which implies that these patients are at increased risk of cardiovascular disease (CVD). Indeed, the prevalence of CVD in MASLD patients is increased and CVD is recognized as the most common cause of death in MASLD patients. We here present an update on the pathophysiology of CVD in MASLD, discuss the risk factors, and suggest screening for CVD in patients with MASLD. Currently, there is no FDA-approved pharmacological treatment for MASLD, and no specific treatment recommended for CVD in patients with MASLD. Thus, the treatment strategy is based on weight loss and a reduction and treatment of CVD risk factors. We recommend screening of MASLD patients for CVD using the SCORE2 system with guidance to specific treatment algorithms. In all patients with CVD risk factors, lifestyle intervention to induce weight loss through diet and exercise is recommended. Especially a Mediterranean diet may improve hyperlipidemia and if further treatment is needed, statins should be used as first-line treatment. Further, anti-hypertensive drugs should be used to treat hypertension. With the epidemic of obesity and type 2 diabetes mellitus (T2DM) the risk of MASLD and CVD is expected to increase, and preventive measures, screening, and effective treatments are highly needed to reduce morbidity and mortality in MASLD patients.

Application and research progress of magnetic resonance proton density fat fraction in metabolic dysfunction-associated steatotic liver disease: a comprehensive review.

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Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD), formerly known as Non-Alcoholic Fatty Liver Disease (NAFLD), is a chronic liver disorder associated with disturbances in lipid metabolism. The disease is prevalent worldwide, particularly closely linked with metabolic syndromes such as obesity and diabetes. Magnetic Resonance Proton Density Fat Fraction (MRI-PDFF), serving as a non-invasive and highly quantitative imaging assessment tool, holds promising applications in the diagnosis and research of MASLD. This paper aims to comprehensively review and summarize the applications and research progress of MRI-PDFF technology in MASLD, analyze its strengths and challenges, and anticipate its future developments in clinical practice.

Sequential SAFE Score and Transient Elastography for Detecting Significant Fibrosis in Asian Patients with MASLD.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly known as nonalcoholic fatty liver disease (NAFLD),¹ represents a global public health issue. Fibrosis stage is the most important risk for long-term undesirable outcomes.^{2,3} From recent meta-analyses, all-cause and liver-related mortalities significantly increased from fibrosis stage 2 (significant fibrosis; $F \geq 2$) onward.^{4,5} In primary care setting, those with $F \geq 2$ should be referred to hepatologists; therefore, noninvasive tests to stratify risk of patients with MASLD are crucial. Steatosis-associated fibrosis estimator (SAFE) was recently developed to predict $F \geq 2$.⁶ SAFE has been externally validated and outperformed fibrosis-4 (FIB-4) and

NAFLD fibrosis score (NFS).^{7,8} Recently, international guidelines proposed sequential diagnostic steps, initially using FIB-4 and then transient elastography (TE) in non-low-risk patients.^{9,10} However, the guidelines focused on identifying advanced fibrosis ($F \geq 3$), which might be too late. This study aimed to compare the performance among SAFE, FIB-4, and NFS, and evaluate SAFE-TE sequential approach. We hypothesized that by initially using SAFE, the proportion of patients misclassified as low risk despite already having $F \geq 2$ could be diminished.
